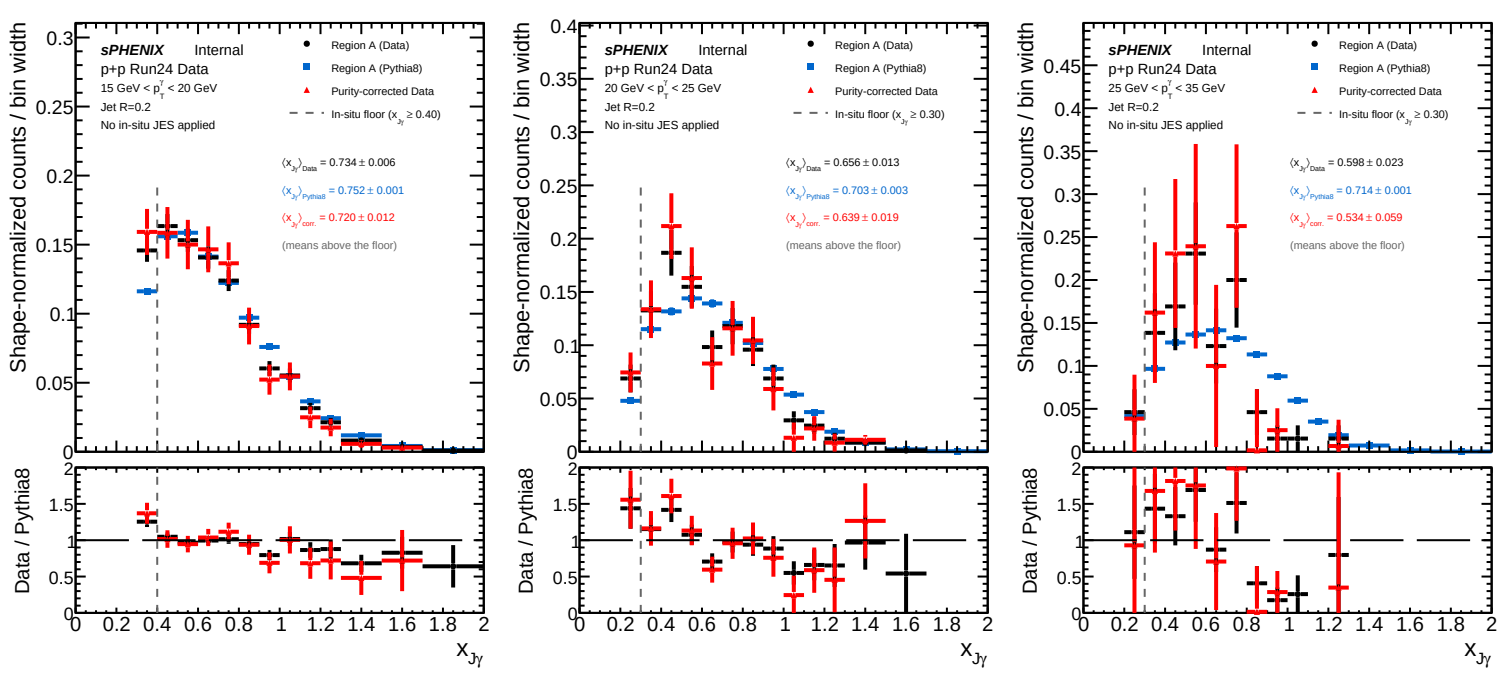
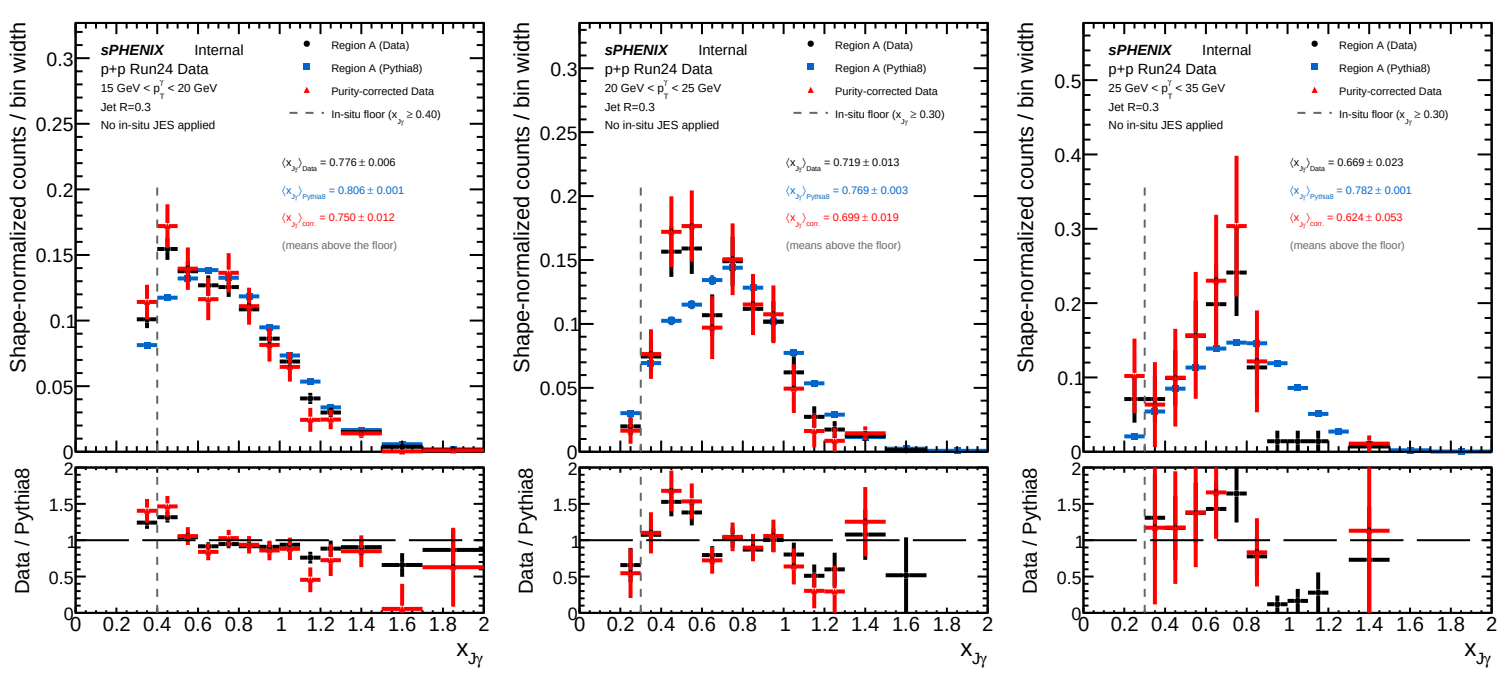
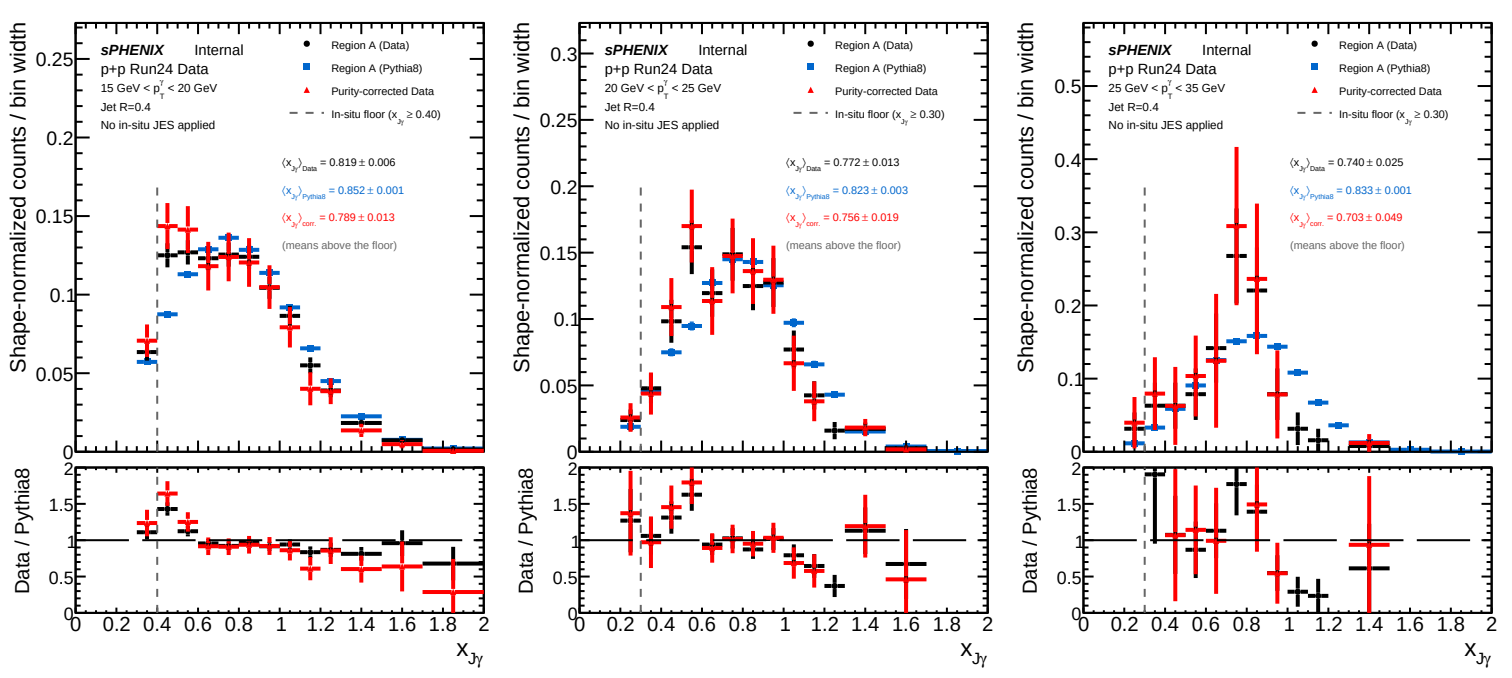
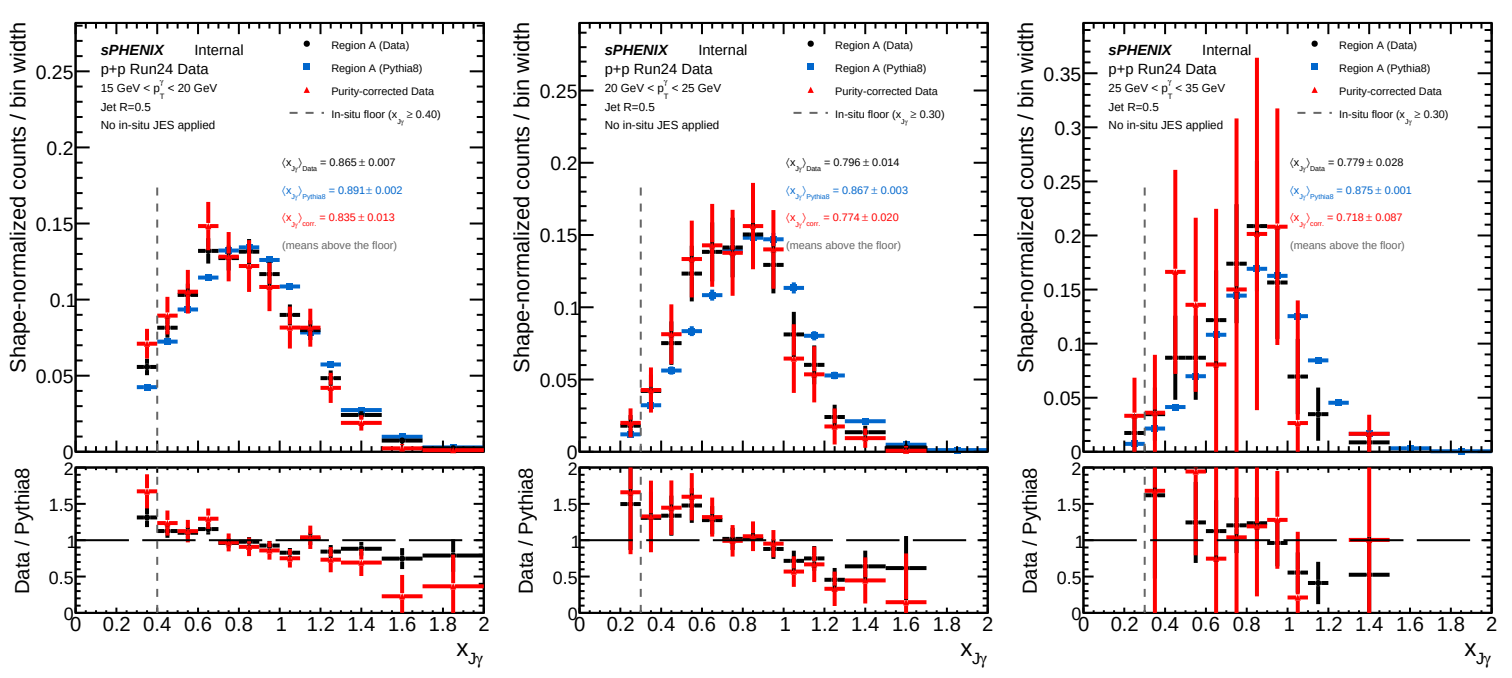
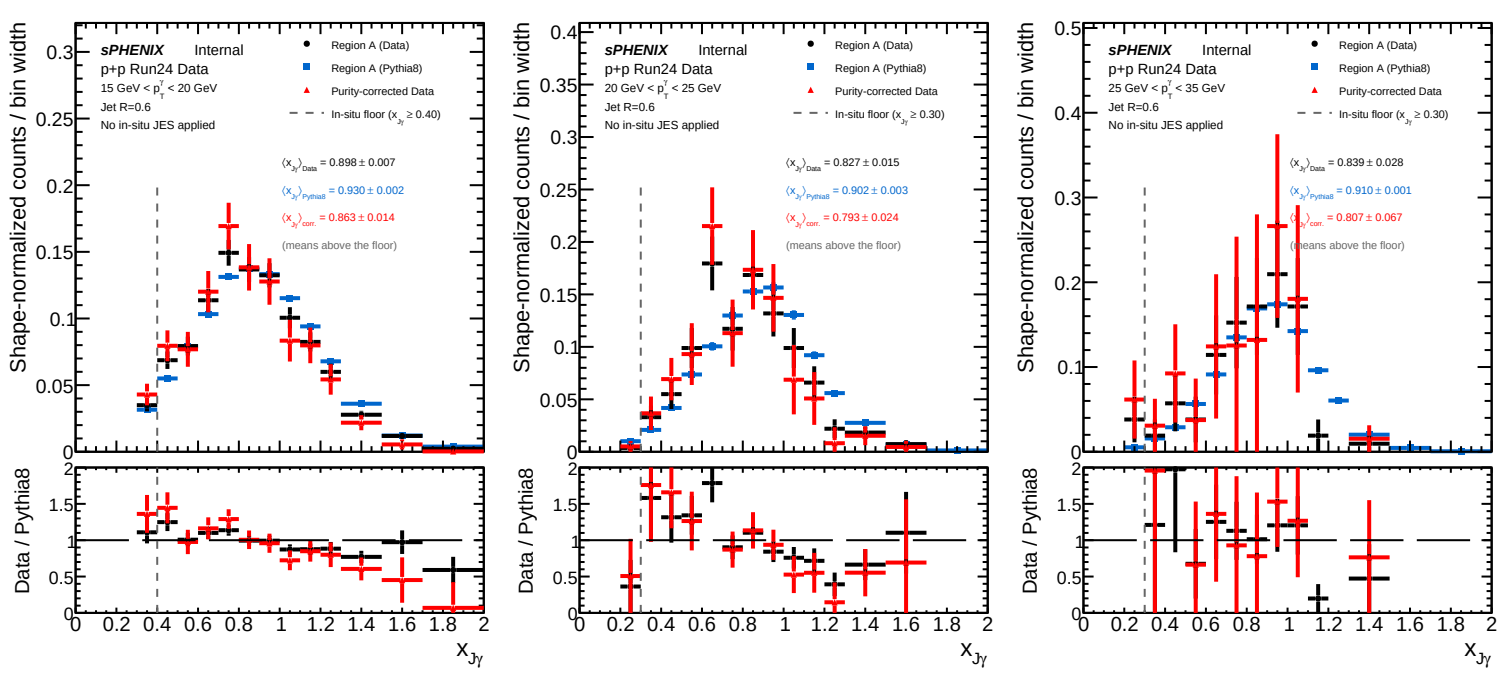
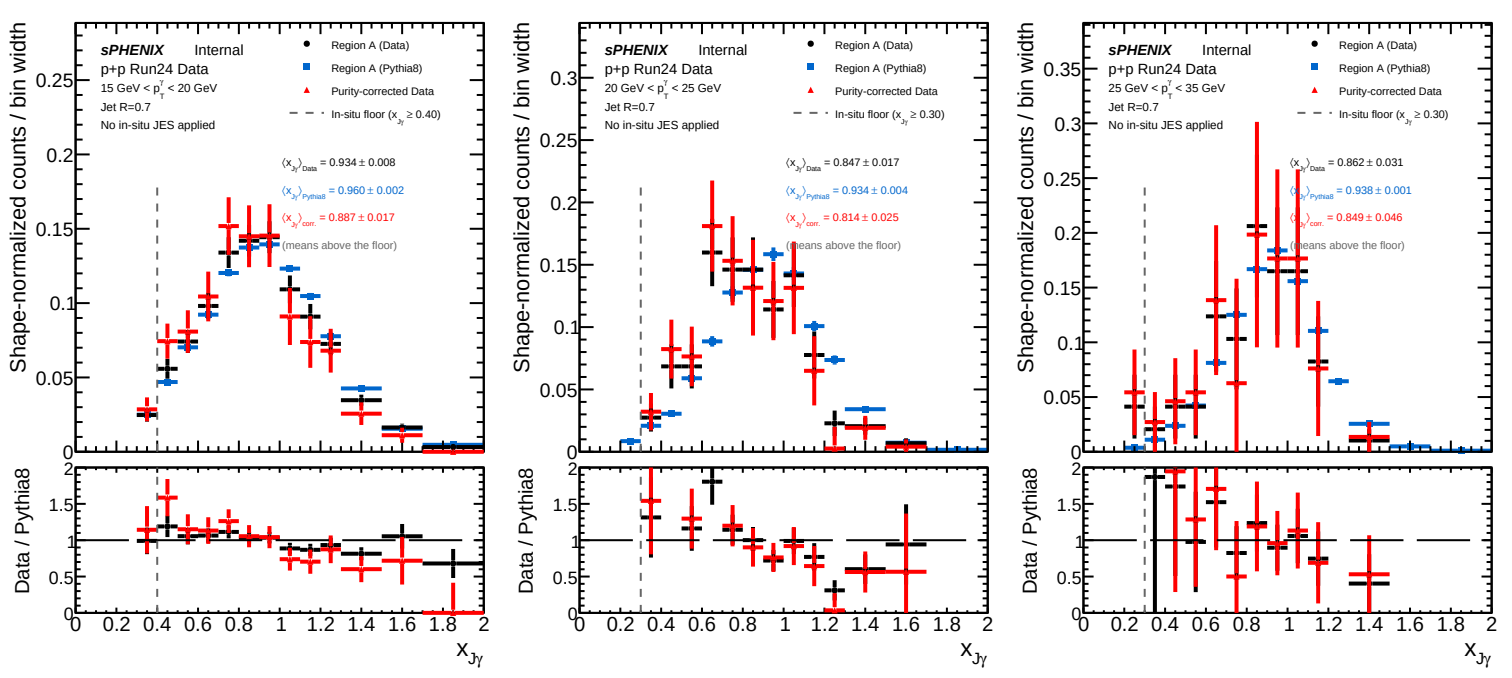

sPHENIX Internal
p+p Run24 Data
15 GeV < p_T^γ < 20 GeV
Jet R=0.2
No in-situ JES applied
Region A (Data)
Region A (Pythia8)
Purity-corrected Data
In-situ floor ($x_{J\gamma} \geq 0.40$)
$\langle x_{J\gamma} \rangle_{Data} = 0.734 \pm 0.006$
$\langle x_{J\gamma} \rangle_{Pythia8} = 0.752 \pm 0.001$
$\langle x_{J\gamma} \rangle_{corr.} = 0.720 \pm 0.012$
(means above the floor)
Shape-normalized counts / bin width
Data / Pythia8
$x_{J\gamma}$
sPHENIX Internal
p+p Run24 Data
20 GeV < p_T^γ < 25 GeV
Jet R=0.2
No in-situ JES applied
Region A (Data)
Region A (Pythia8)
Purity-corrected Data
In-situ floor ($x_{J\gamma} \geq 0.30$)
$\langle x_{J\gamma} \rangle_{Data} = 0.656 \pm 0.013$
$\langle x_{J\gamma} \rangle_{Pythia8} = 0.703 \pm 0.003$
$\langle x_{J\gamma} \rangle_{corr.} = 0.639 \pm 0.019$
(means above the floor)
Shape-normalized counts / bin width
Data / Pythia8
$x_{J\gamma}$
sPHENIX Internal
p+p Run24 Data
25 GeV < p_T^γ < 35 GeV
Jet R=0.2
No in-situ JES applied
Region A (Data)
Region A (Pythia8)
Purity-corrected Data
In-situ floor ($x_{J\gamma} \geq 0.30$)
$\langle x_{J\gamma} \rangle_{Data} = 0.598 \pm 0.023$
$\langle x_{J\gamma} \rangle_{Pythia8} = 0.714 \pm 0.001$
$\langle x_{J\gamma} \rangle_{corr.} = 0.534 \pm 0.059$
(means above the floor)
Shape-normalized counts / bin width
Data / Pythia8
$x_{J\gamma}$

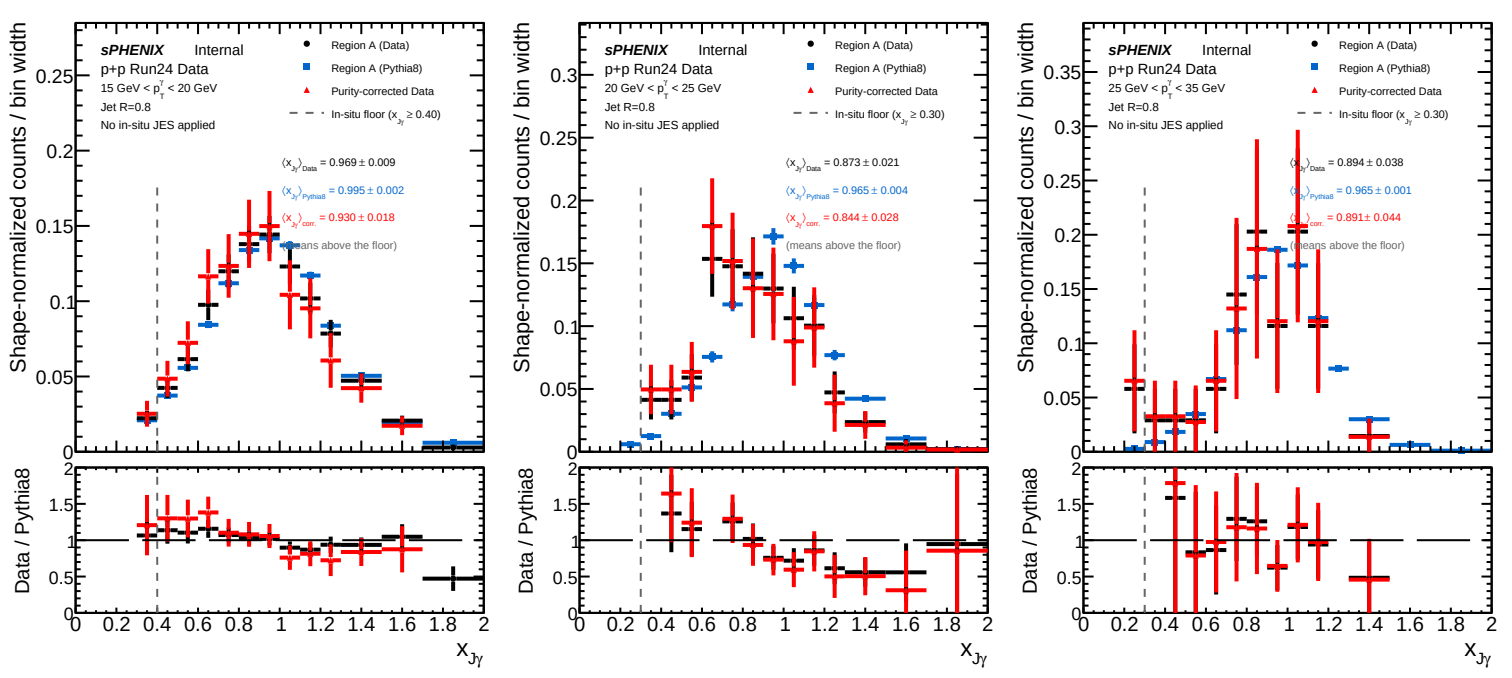

sPHENIX Internal
p+p Run24 Data
15 GeV < p_T^γ < 20 GeV
Jet R=0.3
No in-situ JES applied
Region A (Data)
Region A (Pythia8)
Purity-corrected Data
In-situ floor ($x_{J\gamma} \geq 0.40$)
$\langle x_{J\gamma} \rangle_{Data} = 0.776 \pm 0.006$
$\langle x_{J\gamma} \rangle_{Pythia8} = 0.806 \pm 0.001$
$\langle x_{J\gamma} \rangle_{corr.} = 0.750 \pm 0.012$
(means above the floor)
Shape-normalized counts / bin width
Data / Pythia8
$x_{J\gamma}$
sPHENIX Internal
p+p Run24 Data
20 GeV < p_T^γ < 25 GeV
Jet R=0.3
No in-situ JES applied
Region A (Data)
Region A (Pythia8)
Purity-corrected Data
In-situ floor ($x_{J\gamma} \geq 0.30$)
$\langle x_{J\gamma} \rangle_{Data} = 0.719 \pm 0.013$
$\langle x_{J\gamma} \rangle_{Pythia8} = 0.769 \pm 0.003$
$\langle x_{J\gamma} \rangle_{corr.} = 0.699 \pm 0.019$
(means above the floor)
Shape-normalized counts / bin width
Data / Pythia8
$x_{J\gamma}$
sPHENIX Internal
p+p Run24 Data
25 GeV < p_T^γ < 35 GeV
Jet R=0.3
No in-situ JES applied
Region A (Data)
Region A (Pythia8)
Purity-corrected Data
In-situ floor ($x_{J\gamma} \geq 0.30$)
$\langle x_{J\gamma} \rangle_{Data} = 0.669 \pm 0.023$
$\langle x_{J\gamma} \rangle_{Pythia8} = 0.782 \pm 0.001$
$\langle x_{J\gamma} \rangle_{corr.} = 0.624 \pm 0.053$
(means above the floor)
Shape-normalized counts / bin width
Data / Pythia8
$x_{J\gamma}$


sPHENIX Internal
p+p Run24 Data
15 GeV < p_T^γ < 20 GeV
Jet R=0.4
No in-situ JES applied
Region A (Data)
Region A (Pythia8)
Purity-corrected Data
In-situ floor ($x_{J\gamma} \geq 0.40$)
$\langle x_{J\gamma} \rangle_{Data} = 0.819 \pm 0.006$
$\langle x_{J\gamma} \rangle_{Pythia8} = 0.852 \pm 0.001$
$\langle x_{J\gamma} \rangle_{corr.} = 0.789 \pm 0.013$
(means above the floor)
Shape-normalized counts / bin width
Data / Pythia8
$x_{J\gamma}$
sPHENIX Internal
p+p Run24 Data
20 GeV < p_T^γ < 25 GeV
Jet R=0.4
No in-situ JES applied
Region A (Data)
Region A (Pythia8)
Purity-corrected Data
In-situ floor ($x_{J\gamma} \geq 0.30$)
$\langle x_{J\gamma} \rangle_{Data} = 0.772 \pm 0.013$
$\langle x_{J\gamma} \rangle_{Pythia8} = 0.823 \pm 0.003$
$\langle x_{J\gamma} \rangle_{corr.} = 0.756 \pm 0.019$
(means above the floor)
Shape-normalized counts / bin width
Data / Pythia8
$x_{J\gamma}$
sPHENIX Internal
p+p Run24 Data
25 GeV < p_T^γ < 35 GeV
Jet R=0.4
No in-situ JES applied
Region A (Data)
Region A (Pythia8)
Purity-corrected Data
In-situ floor ($x_{J\gamma} \geq 0.30$)
$\langle x_{J\gamma} \rangle_{Data} = 0.740 \pm 0.025$
$\langle x_{J\gamma} \rangle_{Pythia8} = 0.833 \pm 0.001$
$\langle x_{J\gamma} \rangle_{corr.} = 0.703 \pm 0.049$
(means above the floor)
Shape-normalized counts / bin width
Data / Pythia8
$x_{J\gamma}$


sPHENIX Internal
p+p Run24 Data
15 GeV < p_T^γ < 20 GeV
Jet R=0.5
No in-situ JES applied
Region A (Data)
Region A (Pythia8)
Purity-corrected Data
In-situ floor ($x_{J\gamma} \geq 0.40$)
$\langle x_{J\gamma}\rangle_{Data}$ = 0.865 ± 0.007
$\langle x_{J\gamma}\rangle_{Pythia8}$ = 0.891 ± 0.002
$\langle x_{J\gamma}\rangle_{corr.}$ = 0.835 ± 0.013
(means above the floor)
Shape-normalized counts / bin width
Data / Pythia8
$x_{J\gamma}$
sPHENIX Internal
p+p Run24 Data
20 GeV < p_T^γ < 25 GeV
Jet R=0.5
No in-situ JES applied
Region A (Data)
Region A (Pythia8)
Purity-corrected Data
In-situ floor ($x_{J\gamma} \geq 0.30$)
$\langle x_{J\gamma}\rangle_{Data}$ = 0.796 ± 0.014
$\langle x_{J\gamma}\rangle_{Pythia8}$ = 0.867 ± 0.003
$\langle x_{J\gamma}\rangle_{corr.}$ = 0.774 ± 0.020
(means above the floor)
Shape-normalized counts / bin width
Data / Pythia8
$x_{J\gamma}$
sPHENIX Internal
p+p Run24 Data
25 GeV < p_T^γ < 35 GeV
Jet R=0.5
No in-situ JES applied
Region A (Data)
Region A (Pythia8)
Purity-corrected Data
In-situ floor ($x_{J\gamma} \geq 0.30$)
$\langle x_{J\gamma}\rangle_{Data}$ = 0.779 ± 0.028
$\langle x_{J\gamma}\rangle_{Pythia8}$ = 0.875 ± 0.001
$\langle x_{J\gamma}\rangle_{corr.}$ = 0.718 ± 0.087
(means above the floor)
Shape-normalized counts / bin width
Data / Pythia8
$x_{J\gamma}$


sPHENIX Internal
p+p Run24 Data
15 GeV < p_T^γ < 20 GeV
Jet R=0.6
No in-situ JES applied
Region A (Data)
Region A (Pythia8)
Purity-corrected Data
In-situ floor ($x_{J\gamma} \geq 0.40$)
$\langle x_{J\gamma}\rangle_{Data} = 0.898 \pm 0.007$
$\langle x_{J\gamma}\rangle_{Pythia8} = 0.930 \pm 0.002$
$\langle x_{J\gamma}\rangle_{corr.} = 0.863 \pm 0.014$
(means above the floor)
Shape-normalized counts / bin width
Data / Pythia8
$x_{J\gamma}$
sPHENIX Internal
p+p Run24 Data
20 GeV < p_T^γ < 25 GeV
Jet R=0.6
No in-situ JES applied
Region A (Data)
Region A (Pythia8)
Purity-corrected Data
In-situ floor ($x_{J\gamma} \geq 0.30$)
$\langle x_{J\gamma}\rangle_{Data} = 0.827 \pm 0.015$
$\langle x_{J\gamma}\rangle_{Pythia8} = 0.902 \pm 0.003$
$\langle x_{J\gamma}\rangle_{corr.} = 0.793 \pm 0.024$
(means above the floor)
Shape-normalized counts / bin width
Data / Pythia8
$x_{J\gamma}$
sPHENIX Internal
p+p Run24 Data
25 GeV < p_T^γ < 35 GeV
Jet R=0.6
No in-situ JES applied
Region A (Data)
Region A (Pythia8)
Purity-corrected Data
In-situ floor ($x_{J\gamma} \geq 0.30$)
$\langle x_{J\gamma}\rangle_{Data} = 0.839 \pm 0.028$
$\langle x_{J\gamma}\rangle_{Pythia8} = 0.910 \pm 0.001$
$\langle x_{J\gamma}\rangle_{corr.} = 0.807 \pm 0.067$
(means above the floor)
Shape-normalized counts / bin width
Data / Pythia8
$x_{J\gamma}$


sPHENIX Internal
p+p Run24 Data
15 GeV < p_T^γ < 20 GeV
Jet R=0.7
No in-situ JES applied
Region A (Data)
Region A (Pythia8)
Purity-corrected Data
In-situ floor ($x_{J\gamma} \geq 0.40$)
$\langle x_{J\gamma}\rangle_{Data} = 0.934 \pm 0.008$
$\langle x_{J\gamma}\rangle_{Pythia8} = 0.960 \pm 0.002$
$\langle x_{J\gamma}\rangle_{corr.} = 0.887 \pm 0.017$
(means above the floor)
Shape-normalized counts / bin width
Data / Pythia8
$x_{J\gamma}$
sPHENIX Internal
p+p Run24 Data
20 GeV < p_T^γ < 25 GeV
Jet R=0.7
No in-situ JES applied
Region A (Data)
Region A (Pythia8)
Purity-corrected Data
In-situ floor ($x_{J\gamma} \geq 0.30$)
$\langle x_{J\gamma}\rangle_{Data} = 0.847 \pm 0.017$
$\langle x_{J\gamma}\rangle_{Pythia8} = 0.934 \pm 0.004$
$\langle x_{J\gamma}\rangle_{corr.} = 0.814 \pm 0.025$
(means above the floor)
Shape-normalized counts / bin width
Data / Pythia8
$x_{J\gamma}$
sPHENIX Internal
p+p Run24 Data
25 GeV < p_T^γ < 35 GeV
Jet R=0.7
No in-situ JES applied
Region A (Data)
Region A (Pythia8)
Purity-corrected Data
In-situ floor ($x_{J\gamma} \geq 0.30$)
$\langle x_{J\gamma}\rangle_{Data} = 0.862 \pm 0.031$
$\langle x_{J\gamma}\rangle_{Pythia8} = 0.938 \pm 0.001$
$\langle x_{J\gamma}\rangle_{corr.} = 0.849 \pm 0.046$
(means above the floor)
Shape-normalized counts / bin width
Data / Pythia8
$x_{J\gamma}$


sPHENIX Internal
p+p Run24 Data
15 GeV < p_T^γ < 20 GeV
Jet R=0.8
No in-situ JES applied
Region A (Data)
Region A (Pythia8)
Purity-corrected Data
In-situ floor ($x_{J\gamma} \geq 0.40$)
$\langle x_{J\gamma}\rangle_{Data} = 0.969 \pm 0.009$
$\langle x_{J\gamma}\rangle_{Pythia8} = 0.995 \pm 0.002$
$\langle x_{J\gamma}\rangle_{corr.} = 0.930 \pm 0.018$
(means above the floor)
Shape-normalized counts / bin width
Data / Pythia8
$x_{J\gamma}$
sPHENIX Internal
p+p Run24 Data
20 GeV < p_T^γ < 25 GeV
Jet R=0.8
No in-situ JES applied
Region A (Data)
Region A (Pythia8)
Purity-corrected Data
In-situ floor ($x_{J\gamma} \geq 0.30$)
$\langle x_{J\gamma}\rangle_{Data} = 0.873 \pm 0.021$
$\langle x_{J\gamma}\rangle_{Pythia8} = 0.965 \pm 0.004$
$\langle x_{J\gamma}\rangle_{corr.} = 0.844 \pm 0.028$
(means above the floor)
Shape-normalized counts / bin width
Data / Pythia8
$x_{J\gamma}$
sPHENIX Internal
p+p Run24 Data
25 GeV < p_T^γ < 35 GeV
Jet R=0.8
No in-situ JES applied
Region A (Data)
Region A (Pythia8)
Purity-corrected Data
In-situ floor ($x_{J\gamma} \geq 0.30$)
$\langle x_{J\gamma}\rangle_{Data} = 0.894 \pm 0.038$
$\langle x_{J\gamma}\rangle_{Pythia8} = 0.965 \pm 0.001$
$\langle x_{J\gamma}\rangle_{corr.} = 0.891 \pm 0.044$
(means above the floor)
Shape-normalized counts / bin width
Data / Pythia8
$x_{J\gamma}$